

HAMMER: Homomorphic Analytics with Multi-Party Method for Encrypted Relational Queries

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Motivation: The SPIR Challenge

Goal: Symmetric Private Information Retrieval (SPIR)

Analytical SQL execution where the client's intent is hidden from the server, and the server's records remain private from the client. No single protocol is efficient enough for this bidirectional privacy.

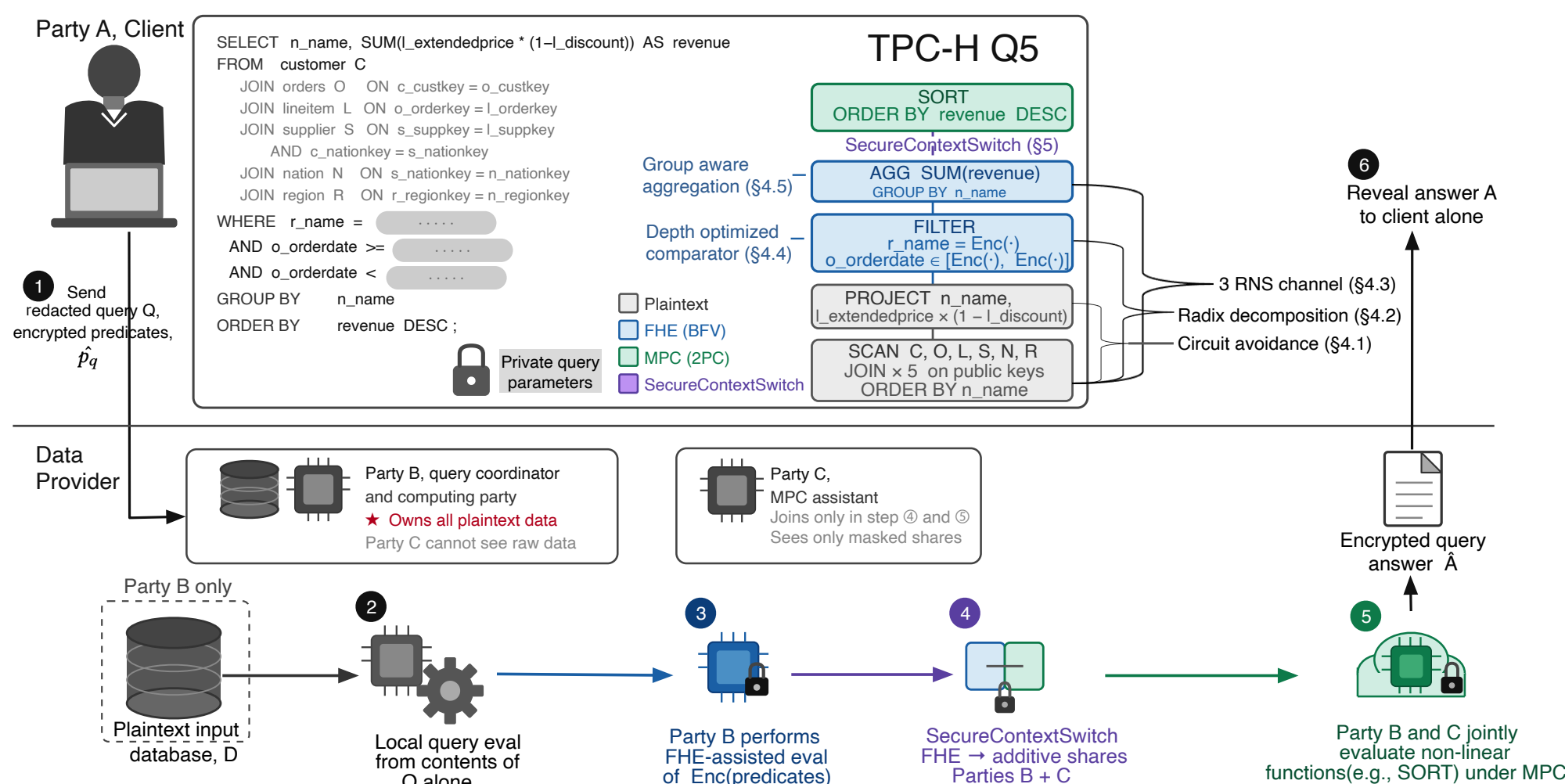
Protocol	Strengths	Bottlenecks
FHE (BFV)	SIMD-parallelism for linear functions.	Deep circuits (sorting) require expensive bootstrapping.
MPC	Efficient complex logic and pipelined gates.	Communication scales with input cardinality.

Hybrid Partitioning

Assign each operator to a domain based on protection needs:

- **Plaintext:** Public structural work (joins, projections).
- **FHE:** Slot-wise linear operators (filters, aggregates) to reduce data size.
- **MPC:** Nonlinear operators on *post-aggregation* state.
- **Result:** Maximizes SIMD-parallelization during FHE and minimizes MPC costs by deferring the handoff until data is reduced to small output groups.

Architecture



Secure Operator Suite

HAMMER utilizes a depth-optimized operator suite to saturate hardware throughput.

(1) Circuit Avoidance

- Encrypts only attributes required for private evaluation.
- Benefit: Shrinks working set to fit in cache and enables fast ct-pt multiplications.

(2) Radix Decomposition

- Large numeric attributes are split into d base- B digits.
- Default Radix-4: Optimized balance between digit count and circuit depth.
- Single-digit encoding is used for small categorical domains.

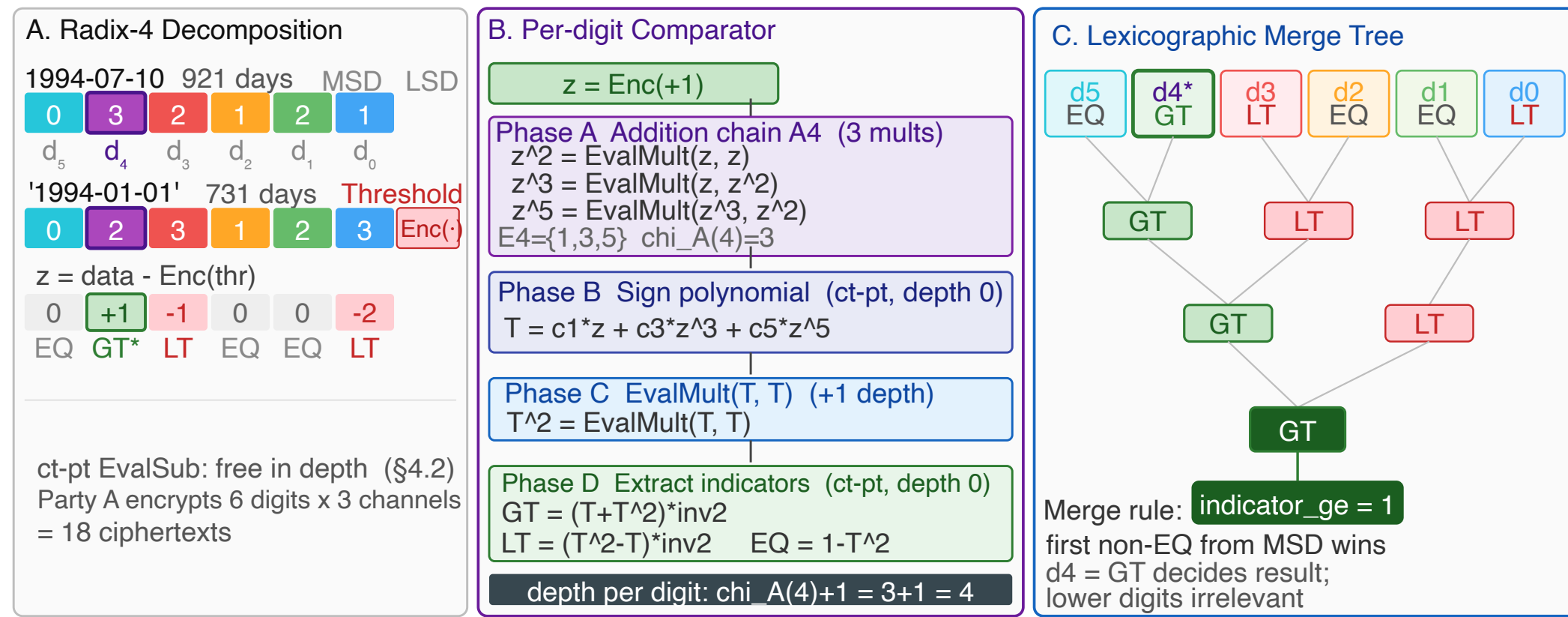
(3) Depth-Optimized BFV Comparator

- Uses a ternary sign polynomial and precomputed addition chains to minimize depth.
- **Impact:** Achieves depth 11–14 with ring dimension $N = 2^{15} - 2^{16}$.

(4) Exact 64-bit Aggregation (Plaintext-RNS)

- Decomposes plaintext modulus into three small coprime channels $\{t_1, t_2, t_3\}$.
- Precision: Product exceeds 2^{64} , ensuring zero precision tradeoff for SQL.
- Performance: Independent channels parallelize across threads/GPU streams.

Comparator Workflow



SecureContextSwitch: FHE → MPC

Input: Party B holds $c_{res} = Enc_{pk}(M)$

Output: $[M]_B = R$, $[M]_C = M - R$

1: **Masking (Party B):**

2: Sample $R \leftarrow \mathbb{Z}_t^{N/2}$

3: $c_{masked} \leftarrow c_{res} - R$

// ct-pt, free in depth

4: Store $[M]_B \leftarrow R$

5: **Distributed Decryption:**

6: $d_B \leftarrow \text{MultipartyDecryptLead}(c_{masked}, s_B)$

7: Send $\{c_{masked}, d_B\}$ to Party C

8: **Fusion (Party C):**

9: $d_C \leftarrow \text{MultipartyDecryptMain}(c_{masked}, s_C)$

10: $y \leftarrow \text{MultipartyFusion}(d_B, d_C)$

// $y = M - R$

11: Store $[M]_C \leftarrow y$

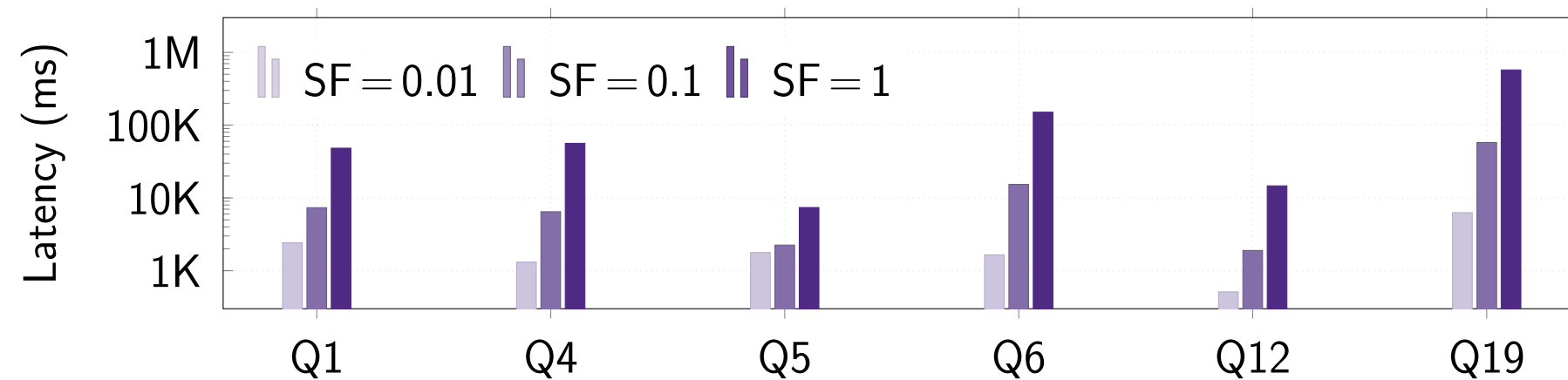
Uniform R makes $M - R$ a one-time pad. Client A offline throughout. Overhead: 1.75% of total query time.

Results

Hammer vs. Engorgio (SF = 0.01, SPR, no GPU).

Q	Hammer (ms)		Eng. (ms)	Speedup	
	Base	Opt		Base	Opt
Q1	17,698	10,350	58,516	$3.31 \times$	$5.65 \times$
Q6	31,786	30,384	160,175	$5.04 \times$	$5.27 \times$
Q12	25,924	22,882	191,689	$7.39 \times$	$8.38 \times$

Scalability across scale factor.



• Hammer vs.Engorgio.

Our hybrid query evaluation achieves $5.3 \times - 8.4 \times$ speedups over the state-of-the-art FHE database on TPC-H workloads.

• Scalability across scale factor.

With GPU-accelerated FHE evaluation, the majority of our query workload runs at SF1 in under a minute.

Contributions

- First SQL system for SPIR with hybrid plaintext/FHE/MPC and client-offline execution.
- SecureContextSwitch: masked FHE-to-MPC handoff with one-time-pad security, post-aggregation.
- Depth-optimized exact BFV operators with 3-channel plaintext-RNS. No bootstrapping.
- Per-query parameter optimizer and GPU acceleration.